(iii) Subtraction of complex numbers

Similarly the difference $z_1 - z_2$ can also be drawn as a position vector whose initial point is the origin and terminal point is $(x_1 - x_2,\ y_1 - y_2)$. We define

$$\begin{aligned} z_1 - z_2 &= z_1 + (-z_2) \\ &= (x_1 + iy_1) + (-x_2 - iy_2) \\ &= (x_1 - x_2) + i(y_1 - y_2). \end{aligned}$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2).$$

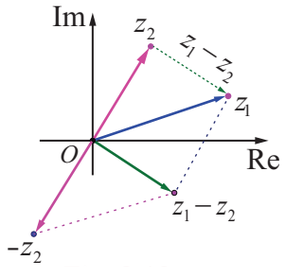


Fig. 2.12

It is important to note here that the vector representing the difference of the vector $z_1 - z_2$ may also be drawn joining the end point of z_2 to the tip of z_1 instead of the origin. This kind of representation does not alter the meaning or interpretation of the difference operator. The difference vector joining the tips of z_1 and z_2 is shown in (green) dotted line.

(iv) Multiplication of complex numbers

The multiplication of complex numbers z_1 and z_2 is defined as

$$\begin{aligned} z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\ &= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1) \end{aligned}$$

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1).$$

Although the product of two complex numbers z_1 and z_2 is itself a complex number represented by a vector, that vector lies in the same plane as the vectors z_1 and z_2. Evidently, then, this product is neither the scalar product nor the vector product used in vector algebra.

Remark

Multiplication of complex number z by i

$$\begin{aligned} \text{If } z &= x + iy, \text{ then} \\ iz &= i(x + iy) \\ &= -y + ix. \end{aligned}$$

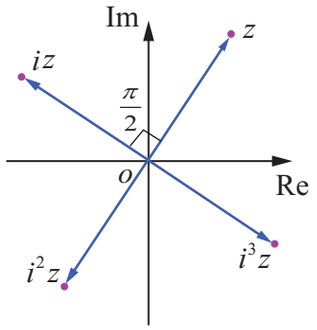


Fig. 2.13

The complex number iz is a rotation of z by 90° or $\frac{\pi}{2}$ radians in the counter clockwise direction about the origin. In general, multiplication of a complex number z by i successively gives a 90° counter clockwise rotation successively about the origin.

Illustration 2.2

Let $z_1 = 6 + 7i$ and $z_2 = 3 - 5i$. Then $z_1 + z_2$ and $z_1 - z_2$ are

(i) $$\begin{aligned} (3 - 5i) + (6 + 7i) &= (3 + 6) + (-5 + 7)i = 9 + 2i \\ (6 + 7i) - (3 - 5i) &= (6 - 3) + (7 - (-5))i = 3 + 12i. \end{aligned}$$

Let $z_1 = 2 + 3i$ and $z_2 = 4 + 7i$. Then $z_1 z_2$ is

(ii) $$\begin{aligned} (2 + 3i)(4 + 7i) &= (2 \times 4 - 3 \times 7) + i(2 \times 7 + 3 \times 4) \\ &= (8 - 21) + (14 + 12)i = -13 + 26i. \end{aligned}$$

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